TOTAL QUALITY MANAGEMENT-23A03705

UNIT - II Historical Review: Quality council, Quality statements, Strategic Planning, Deming Philosophy, Barriers of TQM Implementation, Benefits of TQM, Characteristics of successful quality leader, Contributions of Gurus of TQM, Case studies.

Historical Review of Total Quality Management (TQM)

Introduction

The concept of **quality** has evolved over many decades. Earlier, organizations focused only on **inspection** to remove defective products. Later, they realized that preventing defects is better than detecting them. This led to the development of **Quality Control (QC)**, **Quality Assurance (QA)**, and finally **Total Quality Management (TQM)**.

TQM became popular during the 1980s through the contributions of quality experts such as **W. Edwards Deming, Joseph Juran, Philip Crosby, Kaoru Ishikawa, and Armand Feigenbaum**.

The historical development of TQM also introduced two important organizational elements:

- Quality Council
- Quality Statements

Evolution of Quality Management

Inspection Era

↓

Quality Control (QC)

↓

Statistical Quality Control (SQC)

↓

Quality Assurance (QA)

↓

Total Quality Management (TQM)

↓

Business Excellence

1. Quality Council

Definition

A Quality Council is a group of top executives responsible for establishing quality policies, setting quality objectives, reviewing performance, and ensuring continuous improvement throughout the organization.

Need for a Quality Council

Every organization requires a Quality Council because:

- Provides leadership for quality improvement.
- Aligns quality goals with business goals.
- Promotes customer satisfaction.
- Reviews quality performance.
- Encourages employee participation.
- Allocates resources for improvement projects.

Objectives of Quality Council

The major objectives are:

1. Develop quality policies.
2. Establish quality objectives.
3. Monitor quality improvement activities.
4. Improve customer satisfaction.
5. Reduce defects and costs.
6. Encourage teamwork.
7. Promote continuous improvement.
8. Review quality performance regularly.

Composition of a Quality Council

A Quality Council usually consists of:

CEO / Managing Director

Production Head | Quality Head | HR Manager

Marketing Head | Finance Head | Purchase Head

Customer Representative (Optional)

Members of Quality Council

- Chief Executive Officer (CEO)
- Quality Manager
- Production Manager
- HR Manager
- Marketing Manager
- Finance Manager
- Purchase Manager
- Department Heads

Responsibilities of Quality Council

The council performs several important functions.

1. Develop Quality Vision

Defines where the organization wants to be regarding quality.

Example: "Become the most trusted automobile manufacturer."

2. Establish Quality Policy

Creates policies that guide employees toward quality improvement.

3. Set Quality Objectives

Examples:

- Reduce customer complaints by 20%.
- Achieve 98% customer satisfaction.
- Reduce defects below 1%.

4. Review Quality Performance

The council regularly checks

- Customer complaints
- Product defects
- Process performance
- Delivery performance

5. Allocate Resources

Provides

- Budget
- Equipment

- Training
- Software
- Skilled manpower

6. Promote Continuous Improvement

Encourages

- Kaizen
- Six Sigma
- Lean Manufacturing
- Quality Circles

7. Encourage Employee Participation

Employees are encouraged to

- Suggest improvements
- Solve problems
- Participate in quality teams

8. Review Customer Satisfaction

Measures customer feedback through

- Surveys
- Reviews
- Complaint analysis

Functions of Quality Council

Quality Council

|

| | | |

Policy Objectives Resources Performance

| | | |

Training Improvement Review Customer Focus

Duties of Quality Council

- Form quality improvement teams.
- Conduct quality meetings.

- Approve improvement projects.
- Monitor progress.
- Remove implementation barriers.
- Recognize employee achievements.
- Review customer feedback.
- Encourage innovation.

Benefits of Quality Council

- Better leadership
- Reduced defects
- Improved communication
- Increased customer satisfaction
- Continuous improvement
- Higher productivity
- Lower production costs
- Better teamwork
- Increased employee motivation
- Higher profits

Example

Toyota

Toyota's Quality Council continuously reviews

- Customer complaints
- Vehicle defects
- Supplier quality
- Manufacturing processes

This helped Toyota become one of the world's leading automobile manufacturers.

Advantages

- Better planning
- Strong quality culture
- Improved decision-making
- Faster problem solving
- Higher customer confidence

- Better coordination

Limitations

- Requires commitment from top management.
- Time-consuming.
- Requires trained members.
- Resistance to organizational change.

Diagram: Working of Quality Council

Customer Feedback
↓
Quality Council
↓
Quality Policies
↓
Departments
↓
Improvement Activities
↓
Review & Monitoring
↓
Customer Satisfaction

2. Quality Statements
Definition

Quality Statements describe an organization's commitment to quality and customer satisfaction.

They clearly explain

- What quality means to the organization.
- What the organization wants to achieve.
- How employees should work.

Types of Quality Statements

There are mainly four quality statements.

Quality Statements

Vision Mission Policy Objectives

1. Quality Vision Statement

Definition

A vision statement describes the future position the organization wants to achieve in quality.

Example

"To become the world's most trusted manufacturer by delivering superior quality products."

Characteristics

- Future-oriented
- Inspiring
- Clear
- Easy to understand
- Motivating

2. Quality Mission Statement

Definition

The mission statement explains the organization's purpose and how it will achieve its vision.

Example

"We provide reliable, safe, innovative, and high-quality products that exceed customer expectations."

Characteristics

- Customer focused
- Practical
- Action-oriented
- Clear
- Realistic

3. Quality Policy

Definition

A Quality Policy is an official statement by top management expressing commitment to quality and continuous improvement.

Example

"Our organization is committed to providing products that consistently meet customer requirements through continuous improvement and employee involvement."

Features

- Written by top management

- Customer-oriented
- Easy to understand
- Supports continuous improvement
- Communicated to all employees

Objectives of Quality Policy

- Improve quality
- Meet customer expectations
- Reduce defects
- Improve efficiency
- Enhance employee involvement

4. Quality Objectives

Definition

Quality objectives are measurable goals established to improve quality performance.

Examples

- Reduce defects by 30%.
- Achieve 99% on-time delivery.
- Increase customer satisfaction to 95%.
- Reduce customer complaints by 40%.

Characteristics of Good Quality Objectives

They should be **SMART**.

S Specific

M Measurable

A Achievable

R Relevant

T Time-bound

Relationship among Quality Statements

Vision

↓

Mission

↓

Quality Policy

↓

Quality Objectives

↓

Continuous Improvement

Example of Complete Quality Statements

Vision

Become the leading provider of high-quality products worldwide.

Mission

Deliver innovative, reliable, and customer-focused products through teamwork and continuous improvement.

Quality Policy

We are committed to satisfying customers by delivering defect-free products through employee involvement and continuous improvement.

Quality Objectives

- Customer satisfaction >95%
- Defect rate <1%
- On-time delivery 99%
- Zero customer complaints
- Continuous employee training

Difference Between Vision, Mission, Policy, and Objectives

Vision	**Mission**	**Quality Policy**	**Quality Objectives**
Future dream	Purpose	Commitment	Measurable goals
Long-term	Present focus	Management statement	Numerical targets
Inspirational	Practical	Organizational guideline	Performance measurement
Broad	Action-oriented	Customer focused	Specific

Real-Life Example (Samsung)

Vision

Inspire the world and create the future.

Mission

Deliver innovative products with superior quality.

Quality Policy

Provide defect-free products while continuously improving processes.

Quality Objectives

- Improve customer satisfaction.
- Reduce production defects.
- Improve supplier quality.
- Increase product reliability.

Strategic Planning

In **Total Quality Management (TQM)**, strategic planning integrates **quality into the organization's vision, mission, policies, objectives, and daily operations**. It ensures that every department works toward common quality goals.

Definition

Strategic Planning is a long-term management process used to determine an organization's vision, mission, quality objectives, and action plans to achieve continuous improvement and customer satisfaction.

Objectives of Strategic Planning

The main objectives are:

- Achieve long-term organizational success.
- Improve customer satisfaction.
- Enhance product and service quality.
- Gain competitive advantage.
- Increase productivity and efficiency.
- Reduce costs and defects.
- Encourage continuous improvement.
- Ensure effective utilization of resources.
- Improve employee involvement.
- Increase profitability.

Need for Strategic Planning

Organizations require strategic planning because it:

- Provides a clear direction.
- Aligns all departments with organizational goals.
- Helps respond to market changes.
- Supports quality improvement initiatives.
- Improves customer focus.
- Reduces uncertainty and risks.
- Promotes teamwork and coordination.

Features (Characteristics) of Strategic Planning

- Long-term oriented (typically 3–10 years)
- Customer-focused
- Quality-oriented
- Continuous improvement driven
- Involves top management
- Data-based decision making
- Flexible and adaptable
- Organization-wide participation

Strategic Planning Process

Environmental Analysis
↓
Vision Statement
↓
Mission Statement
↓
Quality Policy & Values
↓
Strategic Quality Objectives
↓
Develop Action Plans
↓
Implementation
↓
Performance Measurement
↓
Continuous Improvement

Steps in Strategic Planning

Step 1: Environmental Analysis

The organization studies internal and external factors affecting its business.

Internal Factors

- Employees
- Technology
- Production capacity
- Financial resources
- Product quality

External Factors

- Customers
- Competitors
- Government regulations
- Market trends
- Economic conditions

Tool Used: SWOT Analysis

SWOT Analysis

SWOT stands for:

Letter	Meaning
S	Strengths
W	Weaknesses
O	Opportunities
T	Threats

Example

Strengths	**Weaknesses**
Skilled employees	Old machinery
Strong brand	High production cost
Opportunities	**Threats**
Growing market	New competitors
New technology	Changing customer preferences

Step 2: Develop Vision
The vision describes what the organization wants to become in the future.
Example "To become the world's leading manufacturer of high-quality products."
Step 3: Develop Mission
The mission explains the organization's purpose.
Example "To provide reliable, innovative, and customer-focused products through continuous improvement."
Step 4: Establish Quality Policy
Quality policy reflects management's commitment to quality.
Example
"We are committed to providing defect-free products that satisfy customer requirements through continuous improvement."
Step 5: Set Strategic Objectives
Objectives should follow the **SMART** principle.

SMART	**Meaning**
S	Specific
M	Measurable
A	Achievable
R	Relevant
T	Time-bound

Examples

- Reduce defects by **30%** within one year.
- Increase customer satisfaction to **95%**.
- Achieve **99%** on-time delivery.
- Reduce production costs by **15%**.

Step 6: Develop Action Plans
The organization prepares detailed plans to achieve objectives.
Action plans include:

- Training employees
- Purchasing new equipment
- Improving production processes
- Implementing Six Sigma or Kaizen
- Supplier quality improvement

Step 7: Implementation
The plans are put into action.
Activities include:

- Employee training
- Teamwork
- Resource allocation
- Process improvement
- Quality monitoring

Step 8: Performance Evaluation
The organization measures performance using Key Performance Indicators (KPIs).
Examples:

- Defect rate
- Customer complaints
- Productivity
- Delivery performance
- Customer satisfaction
- Cost of quality

Step 9: Continuous Improvement
The results are reviewed, and improvements are made continuously.
Common improvement methods:

- Kaizen
- PDCA Cycle
- Six Sigma
- Lean Manufacturing
- Benchmarking

Strategic Planning Cycle

Plan
↓
Implement
↓
Measure
↓
Improve
→ Repeat

Components of Strategic Planning

Strategic Planning

Vision Mission Policy Objectives Action Plans

Role of Top Management
Top management is responsible for:

- Developing the vision and mission
- Establishing quality policies
- Setting quality objectives
- Providing resources
- Encouraging employee participation
- Monitoring performance
- Reviewing strategic plans
- Supporting continuous improvement

Benefits of Strategic Planning

- Clear organizational direction
- Better customer satisfaction
- Improved quality
- Reduced costs

- Higher productivity
- Better resource utilization
- Improved employee motivation
- Better teamwork
- Competitive advantage
- Increased profits
- Continuous improvement

Limitations

- Time-consuming
- Expensive to implement
- Requires strong leadership
- Resistance to change
- Requires continuous monitoring
- Plans may need revision due to changing market conditions

Example: Toyota

Toyota uses strategic planning to:

- Focus on customer satisfaction.
- Reduce manufacturing defects.
- Improve supplier quality.
- Implement **Kaizen (Continuous Improvement)**.
- Invest in employee training.
- Introduce innovative and reliable vehicles.

This long-term approach has helped Toyota maintain a strong global reputation for quality.

Strategic Planning in TQM

Customer Requirements
│
▼
Strategic Planning
│
▼
Quality Objectives
│
▼
Process Improvement
│
▼
High-Quality Products
│
▼
Customer Satisfaction
│
▼
Continuous Improvement

Difference Between Strategic Planning and Operational Planning

Strategic Planning	**Operational Planning**
Long-term	Short-term
Done by top management	Done by middle/lower management
Focuses on overall organization	Focuses on specific departments

Strategic Planning	**Operational Planning**
Sets organizational goals	Implements daily activities
Broad in scope	Detailed and specific
Future-oriented	Present-oriented

Deming Philosophy

Introduction

Dr. W. Edwards Deming is known as the **Father of Modern Quality Management**. He introduced the philosophy that **quality should be built into the process rather than inspected at the end**. His ideas transformed Japanese industries after World War II, making companies like Toyota, Sony, and Honda globally recognized for quality.

Deming believed that **continuous improvement, employee participation, leadership, and customer satisfaction** are the foundations of successful organizations.

About W. Edwards Deming

- **Full Name:** William Edwards Deming
- **Born:** October 14, 1900, USA
- **Profession:** Statistician, Engineer, Professor, and Management Consultant
- **Known As:** Father of Modern Quality Management
- **Major Contributions:**
  - Deming's 14 Points for Management
  - PDCA (Plan–Do–Check–Act) Cycle
  - Statistical Process Control (SPC)
  - Continuous Improvement Philosophy

Definition of Deming Philosophy

Deming Philosophy is a management approach that emphasizes continuous improvement, customer satisfaction, leadership, teamwork, and reducing variation in processes to achieve long-term organizational success.

Basic Principles of Deming Philosophy

Deming believed that:

- Quality is everyone's responsibility.
- Prevention is better than inspection.
- Customer satisfaction is the ultimate goal.
- Continuous improvement should never stop.
- Management is responsible for improving the system.
- Employees should be trained and motivated.
- Decisions should be based on data and facts.

Deming's System of Profound Knowledge

Deming's philosophy is based on **four interconnected areas**.

System of Profound Knowledge

Appreciation for a System | Knowledge of Variation | Theory of Knowledge

Psychology

1. Appreciation for a System

- Understand how all departments work together.
- Optimize the entire organization rather than individual departments.

2. Knowledge of Variation

- Recognize that every process has variation.
- Reduce unnecessary variation to improve quality.

3. Theory of Knowledge

- Decisions should be based on data, learning, and experimentation.

4. Psychology

- Understand employee motivation.
- Encourage teamwork and remove fear.

Deming's 14 Principles for Management

These **14 points** are the foundation of Deming's philosophy.

1. Create Constancy of Purpose

Focus on long-term quality improvement instead of short-term profits.

Example: Investing in employee training rather than only increasing production.

2. Adopt the New Philosophy

Accept quality as a way of doing business.

Example: Produce defect-free products instead of fixing defects later.

3. Stop Dependence on Inspection

Build quality into the production process rather than relying on final inspection.

Example: Use process monitoring during manufacturing.

4. End Buying Based on Price Alone

Choose suppliers based on quality and reliability, not only cost.

Example: Purchasing high-quality raw materials even if they are slightly more expensive.

5. Improve Every Process Continuously

Improve products, services, and processes regularly.

Example: Implement Kaizen activities every month.

6. Institute Training

Provide continuous training for employees.

Example: Training workers on new manufacturing technologies.

7. Institute Leadership

Managers should guide, support, and coach employees instead of only supervising them.

Example: Helping employees solve production problems.

8. Drive Out Fear

Employees should feel free to report problems without fear of punishment.

Example: Encourage workers to report defects immediately.

9. Break Down Barriers Between Departments

Promote teamwork and communication among departments.

Example: Production, design, and marketing working together to improve products.

10. Eliminate Slogans and Targets

Avoid unrealistic slogans without providing methods to achieve them.

Example: Instead of saying "Zero Defects," provide training and better equipment.

11. Eliminate Numerical Quotas

Focus on quality rather than only quantity.

Example: Reward employees for producing quality products instead of just meeting production targets.

12. Remove Barriers to Pride in Workmanship

Provide employees with the resources needed to perform quality work.

Example: Maintain machines properly so workers can produce defect-free products.

13. Encourage Education and Self-Improvement

Support employees in improving their skills.

Example: Sponsoring certification programs and technical workshops.

14. Everyone Must Work Toward Transformation

Quality improvement should involve everyone in the organization.

Example: Every department participates in quality improvement teams.

PDCA Cycle (Deming Cycle)

The **PDCA Cycle** is a systematic method for continuous improvement.

```
PLAN
 │
 ▼
DO
 │
 ▼
CHECK
 │
 ▼
ACT
 │
 └──────────────► Repeat
```

1. Plan

- Identify problems.
- Set improvement goals.
- Develop solutions.

2. Do

- Implement the planned changes on a small scale.

3. Check

- Measure and analyze the results.

4. Act

- Standardize successful improvements.
- Begin the next improvement cycle.

Benefits of Deming Philosophy

- Improves product quality.
- Increases customer satisfaction.
- Reduces defects and waste.
- Enhances employee involvement.
- Improves teamwork.
- Encourages continuous improvement.
- Reduces production costs.
- Increases productivity.
- Strengthens competitive advantage.
- Improves organizational performance.

Limitations

- Requires strong management commitment.
- Takes time to show results.
- Employees may resist change.
- Requires continuous training.
- Implementation can be costly initially.

Real-Life Example: Toyota

Toyota successfully applies Deming's philosophy by:

- Practicing **Kaizen (Continuous Improvement)**.
- Using the **PDCA Cycle** to improve processes.
- Training employees regularly.
- Encouraging teamwork and problem-solving.
- Building quality into every manufacturing stage.
- Focusing on customer satisfaction rather than only production volume.

As a result, Toyota is recognized worldwide for producing reliable and high-quality vehicles.

Diagram: Deming Philosophy in TQM

Customer Needs
↓
Leadership
↓
Employee Involvement
↓
Continuous Improvement
↓
High-Quality Products

↓

Customer Satisfaction

↓

Business Success

Deming Philosophy vs Traditional Management

Deming Philosophy	Traditional Management
Focus on prevention	Focus on inspection
Continuous improvement	Correct defects after production
Teamwork	Individual performance
Long-term quality	Short-term profits
Leadership and coaching	Supervision and control
Data-based decisions	Opinion-based decisions
Customer satisfaction	Production targets

Major Barriers to TQM Implementation

1. Lack of Top Management Commitment

Top management plays a crucial role in implementing TQM. Without their support, quality initiatives often fail.

Problems

- Insufficient funding
- No clear quality vision
- Weak leadership
- Lack of motivation

Example: A company starts a quality improvement program but management does not provide resources or review progress.

2. Resistance to Change

Employees may resist TQM because they fear:

- New responsibilities
- Job insecurity
- Increased workload
- Changes in work methods

Example: Workers refuse to adopt new quality procedures because they are comfortable with existing methods.

3. Poor Employee Involvement

TQM requires participation from everyone.

If employees are not involved:

- Improvement ideas decrease.
- Teamwork becomes weak.
- Motivation reduces.

Example: Management makes all decisions without consulting employees.

4. Lack of Training

Employees need proper training in:

- Quality tools
- Statistical techniques
- Problem-solving methods
- Process improvement

Without training, employees cannot effectively implement TQM.

5. Poor Communication

Communication problems create misunderstandings about quality goals.

Effects include:

- Confusion
- Delays
- Errors
- Lack of coordination

6. Inadequate Customer Focus

Some organizations focus only on production instead of customer needs.

This results in:

- Customer dissatisfaction
- Increased complaints
- Loss of market share

7. Insufficient Resources

TQM requires:

- Budget
- Skilled employees
- Modern equipment
- Time

Lack of these resources slows implementation.

8. Poor Organizational Culture

A culture that discourages innovation or teamwork becomes a major obstacle.

Examples:

- Employees are afraid to report mistakes.
- Departments do not cooperate.

9. Lack of Continuous Improvement

Some organizations treat TQM as a one-time project instead of an ongoing process.

This causes quality improvements to stop after initial success.

10. Poor Planning

Without proper planning:

- Goals become unclear.
- Resources are wasted.
- Projects fail.

11. Inadequate Measurement

Organizations must measure quality using indicators such as:

- Defect rate
- Customer satisfaction
- Productivity
- Delivery performance

Without measurement, improvement cannot be evaluated.

12. Poor Supplier Quality

Low-quality raw materials lead to poor final products.

Organizations should develop long-term relationships with reliable suppliers.

Diagram: Barriers to TQM

Barriers to TQM

Management | Employees | Resources | Organization

No Support | Resistance | Lack of Money | Poor Culture
Training | Poor Teamwork | Old Equipment | Poor Planning
Communication | Customer Focus | Measurement Problems

Summary of Barriers

Barrier	Effect
Lack of management support	Weak implementation
Resistance to change	Slow improvement
Poor employee involvement	Low motivation
Lack of training	Poor quality performance
Poor communication	Misunderstandings
Lack of customer focus	Customer dissatisfaction
Insufficient resources	Delayed implementation
Poor organizational culture	Low teamwork
Lack of continuous improvement	No long-term success
Poor planning	Project failure
Poor measurement	Difficult to evaluate performance
Poor supplier quality	Increased defects

Example

A manufacturing company introduces TQM but:

- Employees are not trained.
- Management does not provide support.
- Departments work independently.
- Customer feedback is ignored.

As a result:

- Product defects increase.
- Customer complaints rise.
- The TQM program fails.

Overcoming the Barriers

Organizations can overcome these barriers by:

- Strong top management commitment.
- Employee participation.
- Continuous training.
- Effective communication.
- Customer-focused culture.
- Teamwork.
- Continuous improvement.
- Proper planning and performance measurement.
- Good supplier relationships.

Benefits of Total Quality Management (TQM)

Introduction

When TQM is implemented successfully, it improves the organization's performance, customer satisfaction, and competitiveness.

Definition

Benefits of TQM are the positive outcomes achieved by implementing continuous quality improvement practices throughout the organization.

Major Benefits of TQM

1. Improved Customer Satisfaction

TQM focuses on understanding and meeting customer expectations.

Benefits:

- Better product quality
- Fewer complaints
- Greater customer loyalty

2. Improved Product Quality

Continuous improvement helps produce reliable and defect-free products.

3. Reduced Defects

Process improvements reduce:

- Rework
- Scrap
- Waste
- Errors

4. Increased Productivity

Improved processes allow employees to produce more in less time.

5. Reduced Costs

TQM lowers costs by reducing:

- Defects
- Waste
- Inspection
- Rework

6. Better Employee Involvement

Employees participate in:

- Decision-making
- Problem-solving

- Improvement teams

This increases motivation and job satisfaction.

7. Better Teamwork

Departments cooperate effectively, improving coordination and communication.

8. Continuous Improvement

Organizations regularly improve:

- Products
- Services
- Processes

using methods like **Kaizen** and the **PDCA Cycle**.

9. Improved Decision Making

TQM encourages decisions based on:

- Facts
- Data
- Statistical analysis

rather than assumptions.

10. Competitive Advantage

High-quality products improve market reputation and attract more customers.

11. Better Supplier Relationships

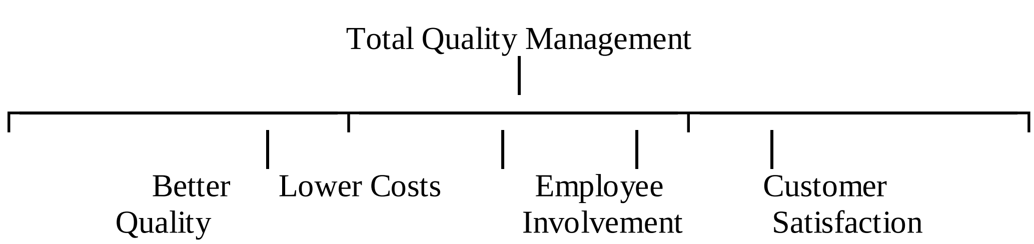
Organizations collaborate with suppliers to improve material quality and reduce defects.

12. Increased Profitability

Improved quality and lower costs lead to:

- Higher sales
- Better customer retention
- Increased profits

Diagram: Benefits of TQM



Higher Productivity Better Teamwork Higher Profits

Summary of Benefits

Benefit	Result
Improved customer satisfaction	Loyal customers
Better product quality	Fewer defects
Reduced costs	Higher profits
Increased productivity	Efficient operations
Employee involvement	Higher motivation
Better teamwork	Improved coordination
Continuous improvement	Long-term success
Better decisions	Improved performance
Competitive advantage	Larger market share
Better supplier relationships	High-quality materials

Real-Life Example: Toyota

Toyota follows TQM by:

- Continuously improving processes (**Kaizen**).
- Training employees regularly.
- Working closely with suppliers.
- Using quality circles to solve problems.
- Focusing on customer satisfaction.

Results:

- High product reliability
- Low defect rates
- Strong global reputation
- Increased customer loyalty
- Higher profitability

Comparison: Barriers vs Benefits of TQM

Barriers to TQM	Benefits of TQM
Lack of management support	Strong leadership
Resistance to change	Continuous improvement
Poor communication	Better teamwork
Lack of training	Skilled employees
Poor customer focus	High customer satisfaction
Resource shortage	Higher productivity
Poor planning	Better organizational performance
Weak supplier relationships	Improved supplier quality

Characteristics of a Successful Quality Leader

Introduction

In **Total Quality Management (TQM)**, leadership is one of the most important principles. A **quality leader** is a person who guides the organization toward achieving high standards of quality, continuous improvement, and customer satisfaction.

A successful quality leader motivates employees, develops a quality culture, and ensures that quality becomes everyone's responsibility.

Definition

A successful quality leader is a person who inspires, guides, and supports employees to achieve organizational goals through continuous improvement, customer focus, teamwork, and effective quality management practices.

Objectives of a Quality Leader

A quality leader aims to:

- Achieve customer satisfaction.
- Improve product and service quality.
- Develop a quality-oriented culture.
- Encourage teamwork.
- Motivate employees.
- Promote continuous improvement.
- Reduce defects and waste.
- Increase organizational performance.

Characteristics of a Successful Quality Leader

1. Customer Focus

A quality leader always gives the highest priority to customers by understanding and meeting their needs.

Example: Collecting customer feedback regularly and improving products based on suggestions.

2. Visionary Leadership

A successful leader has a clear vision for the future and communicates it effectively.

Example

"Become the leading manufacturer of high-quality products."

3. Commitment to Quality

The leader demonstrates a strong commitment to maintaining and improving quality in every activity.

Example: Supporting quality improvement programs and allocating resources.

4. Excellent Communication Skills

A quality leader communicates:

- Goals
- Policies
- Expectations
- Feedback

clearly to all employees.

5. Employee Empowerment

A quality leader trusts employees and encourages them to make decisions and solve problems.

Benefits

- Higher motivation
- Greater responsibility
- Better innovation

6. Teamwork Orientation

Quality leaders encourage cooperation among departments and employees.

Example: Creating quality circles and cross-functional teams.

7. Continuous Improvement Mindset

A successful leader believes that improvement should never stop.

Common methods include:

- Kaizen
- PDCA Cycle
- Six Sigma
- Lean Manufacturing

8. Decision-Making Based on Facts

Quality leaders use:

- Data
- Statistics
- Customer feedback
- Performance reports

instead of assumptions.

9. Integrity and Honesty

A leader should be ethical, honest, and transparent in all decisions.

This builds trust among employees and customers.

10. Problem-Solving Ability

A quality leader identifies root causes of problems and implements permanent solutions.

Example: Using Cause-and-Effect (Fishbone) Diagrams and the 5 Whys technique.

11. Ability to Motivate Employees

A quality leader inspires employees through:

- Recognition
- Rewards
- Encouragement
- Career development

Motivated employees contribute more effectively to quality improvement.

12. Commitment to Training and Learning

Quality leaders ensure continuous learning by organizing:

- Technical training
- Quality workshops
- Skill development programs

13. Adaptability to Change

A successful leader accepts new technologies, methods, and changing customer expectations.

Example: Implementing digital quality management systems.

14. Long-Term Thinking

Quality leaders focus on sustainable success rather than short-term profits.

They invest in:

- Employee development
- Better technology
- Process improvement

15. Accountability

A quality leader takes responsibility for both successes and failures and encourages accountability throughout the organization.

Diagram: Characteristics of a Successful Quality Leader

Roles of a Quality Leader

A quality leader performs the following roles:

- Sets the vision and quality goals.
- Develops quality policies.
- Motivates employees.
- Promotes teamwork.
- Encourages innovation.
- Provides training.
- Solves organizational problems.
- Reviews quality performance.
- Supports continuous improvement.
- Builds customer confidence.

Importance of a Quality Leader

A successful quality leader helps the organization by:

- Improving customer satisfaction.
- Increasing employee motivation.
- Reducing defects.
- Improving communication.
- Building teamwork.
- Increasing productivity.
- Enhancing organizational reputation.
- Achieving long-term success.

Example: Toyota

Toyota's leaders demonstrate quality leadership by:

- Encouraging **Kaizen (Continuous Improvement)**.
- Empowering employees to stop the production line if defects are found (Jidoka).

- Promoting teamwork and problem-solving.
- Investing in employee training.
- Making decisions based on data and customer feedback.

Result:

- High-quality products
- Low defect rates
- Strong customer trust
- Global market leadership

Advantages of Effective Quality Leadership

- Better customer satisfaction
- Improved product quality
- Higher employee morale
- Better teamwork
- Increased productivity
- Reduced waste and defects
- Improved organizational performance
- Strong competitive advantage
- Continuous improvement culture
- Higher profitability

Qualities of a Traditional Manager vs. Quality Leader

Traditional Manager	Successful Quality Leader
Focuses on control	Focuses on guidance and coaching
Gives orders	Inspires employees
Short-term goals	Long-term vision
Individual performance	Teamwork
Corrects mistakes	Prevents mistakes
Decision based on authority	Decision based on facts and data
Limited employee involvement	Employee empowerment

Summary (Key Characteristics)

A successful quality leader should possess:

- Customer focus
- Visionary thinking
- Commitment to quality
- Effective communication
- Employee empowerment
- Teamwork
- Continuous improvement mindset
- Data-based decision making
- Integrity and honesty
- Problem-solving ability
- Motivation skills

- Commitment to training
- Adaptability
- Long-term thinking
- Accountability

Contributions of Gurus of TQM

Introduction

The development of **Total Quality Management (TQM)** is largely due to the contributions of several quality experts, known as the **Quality Gurus**. These experts introduced principles, techniques, and management philosophies that help organizations improve quality, reduce defects, and achieve customer satisfaction.

The most important TQM gurus are:

1. **W. Edwards Deming**
2. **Joseph M. Juran**
3. **Philip B. Crosby**
4. **Kaoru Ishikawa**
5. **Armand V. Feigenbaum**
6. **Genichi Taguchi**
7. **Shigeo Shingo**

1. W. Edwards Deming (1900–1993)

Introduction

Deming is known as the **Father of Modern Quality Management**. He helped Japanese industries improve quality after World War II through statistical quality control and continuous improvement.

Major Contributions

1. Deming's 14 Points for Management

These 14 principles help organizations improve quality and productivity.

Examples include:

- Create constancy of purpose.
- Drive out fear.
- Improve processes continuously.
- Remove barriers between departments.
- Encourage education and training.

2. PDCA Cycle (Deming Cycle)

The **Plan–Do–Check–Act (PDCA)** cycle is used for continuous improvement.

PLAN
↓
DO
↓
CHECK
↓
ACT
└──► Repeat

3. Statistical Process Control (SPC)

Deming promoted the use of statistical methods to control and improve manufacturing processes.

4. Continuous Improvement

He emphasized that quality improvement should never stop.

Key Idea

"Quality should be built into the process rather than inspected at the end."

2. Joseph M. Juran (1904–2008)

Introduction

Joseph Juran emphasized **quality planning and management responsibility**. He believed quality begins with understanding customer needs.

Major Contributions

1. Juran Trilogy

The Juran Trilogy consists of:

Juran Trilogy

Planning Control Improvement

Quality Planning

Identify customer needs and design products accordingly.

Quality Control

Measure actual performance and compare it with standards.

Quality Improvement

Continuously improve processes to eliminate defects.

2. Pareto Principle (80:20 Rule)

Juran popularized the **Pareto Principle**, which states:

80% of problems are caused by 20% of the causes.

Example:

If 100 customer complaints are received, around **80 complaints may be due to only 20 major causes**.

Key Idea

Quality does not happen by accident; it must be planned.

3. Philip B. Crosby (1926–2001)

Introduction

Philip Crosby believed that **quality means conformance to requirements** and that preventing defects is better than correcting them.

Major Contributions

1. Four Absolutes of Quality

- Quality means conformance to requirements.
- Prevention is the system of quality.
- Zero defects is the performance standard.
- Quality is measured by the cost of non-conformance.

2. Zero Defects Concept

Crosby believed organizations should aim for **zero defects**, not "acceptable" defect levels.

3. Cost of Quality

He classified quality costs into:

- Prevention costs
- Appraisal costs
- Internal failure costs
- External failure costs

Key Idea

"Do it right the first time."

4. Kaoru Ishikawa (1915–1989)

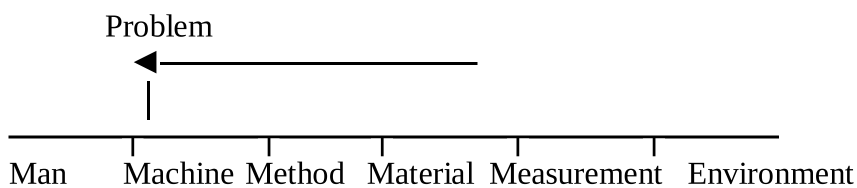
Introduction

Kaoru Ishikawa promoted **company-wide quality control** and employee participation.

Major Contributions

1. Cause-and-Effect Diagram (Fishbone Diagram)

Used to identify the root causes of problems.



2. Quality Circles

Small groups of employees meet regularly to solve work-related problems.

3. Seven Basic Quality Tools

Ishikawa promoted:

- Check Sheet
- Histogram
- Pareto Chart
- Fishbone Diagram
- Scatter Diagram
- Control Chart
- Flow Chart

Key Idea

Every employee should participate in quality improvement.

5. Armand V. Feigenbaum (1922–2014)

Introduction

Feigenbaum introduced the concept of **Total Quality Control (TQC)**.

Major Contributions

1. Total Quality Control

Quality is everyone's responsibility—not just the quality department.

2. Company-Wide Quality

Every department should contribute to quality.

Departments include:

- Production
- Marketing
- Purchasing
- Finance
- Human Resources

3. Cost of Quality

He emphasized measuring the cost associated with poor quality.

Key Idea

Quality is a company-wide responsibility.

6. Genichi Taguchi (1924–2012)

Introduction

Taguchi developed methods to improve product design and reduce variation.

Major Contributions

1. Taguchi Loss Function

Any deviation from the target value causes a financial loss to society.

2. Robust Design

Products should perform consistently even under varying conditions.

3. Design of Experiments (DOE)

Use statistical experiments to determine the best product design and process settings.

Key Idea

Quality should be designed into the product.

7. Shigeo Shingo (1909–1990)

Introduction

Shingo made major contributions to manufacturing quality and productivity.

Major Contributions

1. Poka-Yoke (Mistake Proofing)

Design processes so that mistakes are prevented or detected immediately.

Example: A USB plug fits in only one orientation.

2. Single-Minute Exchange of Dies (SMED)

A method to reduce machine setup time, increasing productivity and flexibility.

3. Lean Manufacturing Support

Shingo's methods reduced waste and improved efficiency.

Key Idea

Prevent mistakes before they become defects.

Comparison of TQM Gurus

Guru	Major Contribution	Key Concept
W. Edwards Deming	14 Points, PDCA Cycle, SPC	Continuous Improvement
Joseph M. Juran	Juran Trilogy, Pareto Principle	Quality Planning
Philip B. Crosby	Zero Defects, Four Absolutes	Do It Right the First Time
Kaoru Ishikawa	Fishbone Diagram, Quality Circles, Seven QC Tools	Employee Participation
Armand V. Feigenbaum	Total Quality Control	Company-Wide Quality
Genichi Taguchi	Loss Function, Robust Design, DOE	Design Quality
Shigeo Shingo	Poka-Yoke, SMED	Mistake Proofing

Importance of TQM Gurus

Their contributions helped organizations to:

- Improve product quality
- Increase customer satisfaction
- Reduce defects
- Reduce production costs
- Improve productivity
- Encourage teamwork
- Promote continuous improvement
- Strengthen global competitiveness

Real-Life Example: Toyota

Toyota applies the ideas of several TQM gurus:

- **Deming:** PDCA Cycle and Continuous Improvement (Kaizen)
- **Juran:** Quality Planning and Improvement
- **Crosby:** Defect Prevention
- **Ishikawa:** Fishbone Diagram and Quality Circles
- **Feigenbaum:** Company-wide Quality Responsibility
- **Taguchi:** Robust Product Design
- **Shingo:** Poka-Yoke and SMED

These practices have helped Toyota become one of the world's leading automobile manufacturers.

Memory Trick (Exam Revision)

Guru	Easy Keyword
Deming	**PDCA & 14 Points**
Juran	**Trilogy & Pareto (80:20)**
Crosby	**Zero Defects**
Ishikawa	**Fishbone & Quality Circles**
Feigenbaum	**Total Quality Control**
Taguchi	**Robust Design & Loss Function**
Shingo	**Poka-Yoke & SMED**

Case Studies

Introduction

A **case study** in **Total Quality Management (TQM)** explains how an organization successfully applies quality principles to improve products, services, customer satisfaction, and business performance.

Case studies help us understand the practical application of concepts such as:

- Customer focus
- Continuous improvement (Kaizen)
- Employee involvement
- Teamwork
- Leadership
- Process improvement
- Quality tools

Case Study 1: Toyota Motor Corporation (Best TQM Example)

Company Profile

- **Industry:** Automobile Manufacturing
- **Country:** Japan
- **Founded:** 1937

Toyota is recognized worldwide for its excellent quality management system and is considered one of the best examples of successful TQM implementation.

Problems Faced

Initially, Toyota experienced:

- Production inefficiencies
- Material wastage
- Manufacturing defects
- High production costs
- Customer expectations for better quality

TQM Practices Adopted

1. Kaizen (Continuous Improvement)

Toyota encourages every employee to suggest improvements, even small ones.

Example: Rearranging tools on the production line to reduce worker movement and save time.

2. Just-in-Time (JIT)

Raw materials arrive only when needed, reducing inventory costs and waste.

3. Jidoka (Automation with Human Intelligence)

Machines automatically stop when defects occur, preventing defective products from moving to the next stage.

4. Quality Circles

Small groups of employees meet regularly to identify problems and suggest improvements.

5. Employee Empowerment

Workers are authorized to stop the production line if they detect a quality problem.

6. PDCA Cycle

Toyota continuously improves processes using:

- Plan
- Do
- Check
- Act

Results Achieved

- High customer satisfaction
- Reduced defects
- Lower production costs
- Increased productivity
- Better employee morale
- Global reputation for quality

- Higher profits

Toyota Production System

Customer Requirements
↓
Just-In-Time (JIT)
↓
Jidoka
↓
Kaizen (CI)
↓
High-Quality Vehicles
↓
Customer Satisfaction

Case Study 2: Motorola and Six Sigma

Company Profile

- **Industry:** Electronics and Telecommunications
- **Country:** USA

Motorola developed the **Six Sigma** quality improvement methodology in the 1980s.

Problems Faced

- High defect rates
- Increasing customer complaints
- High production costs
- Process variations

TQM Practices Adopted

Six Sigma Methodology

Motorola aimed to reduce defects to **3.4 defects per million opportunities (DPMO)**.

DMAIC Process

Define
↓
Measure
↓
Analyze

↓
Improve
↓
Control

Results

- Significant reduction in defects
- Improved product quality
- Reduced production costs
- Higher customer satisfaction
- Increased profits

Case Study 3: Xerox Corporation

Company Profile

- **Industry:** Office Equipment
- **Country:** USA

During the 1980s, Xerox faced strong competition from Japanese companies.

Problems Faced

- High production costs
- Poor product quality
- Loss of market share
- Customer dissatisfaction

TQM Practices Adopted

Benchmarking

Xerox compared its performance with industry leaders and adopted best practices.

Employee Involvement

Employees participated in quality improvement teams.

Customer Focus

Customer feedback was used to improve products.

Results

- Improved quality
- Reduced manufacturing costs
- Better customer satisfaction
- Increased competitiveness

Benchmarking Process

Identify Best Company
↓
Compare Performance
↓
Identify Gaps
↓
Implement Improvements
↓
Continuous Improvement

Case Study 4: Tata Steel (India)

Company Profile

- **Industry:** Steel Manufacturing
- **Country:** India

Tata Steel is one of India's leading organizations implementing TQM.

TQM Practices

- Employee involvement
- Continuous training
- Quality circles
- Customer feedback
- Safety improvement
- Process optimization

Results

- Better product quality
- Improved productivity
- Reduced wastage
- Increased customer trust
- International quality recognition

Case Study 5: Infosys (India)

Company Profile

- **Industry:** Information Technology
- **Country:** India

Infosys follows TQM principles to improve software quality and customer satisfaction.

TQM Practices

- Continuous process improvement
- Employee skill development
- Customer feedback systems
- ISO quality standards
- Risk management
- Performance measurement

Results

- High customer satisfaction
- Reliable software delivery
- Improved productivity
- Strong global reputation

Comparison of Case Studies

Company	Major TQM Practice	Main Result
Toyota	Kaizen, JIT, Jidoka, Quality Circles	High-quality vehicles, low defects
Motorola	Six Sigma (DMAIC)	Very low defect rates
Xerox	Benchmarking	Improved competitiveness
Tata Steel	Employee involvement, Quality Circles	Higher productivity and quality
Infosys	Process improvement, Customer focus	Better software quality and customer satisfaction

Lessons Learned from These Case Studies

- Customer satisfaction is the primary objective.
- Continuous improvement leads to long-term success.
- Employee involvement improves innovation.
- Strong leadership is essential.
- Quality should be built into the process.
- Data-based decision making improves performance.
- Good supplier relationships enhance product quality.

Advantages of Implementing TQM

- Better product quality
- Reduced defects
- Increased customer satisfaction
- Higher productivity
- Lower costs
- Better teamwork
- Improved employee motivation
- Greater competitive advantage
- Continuous improvement
- Higher profitability